

I Want to be Iron Man: Exploring JARVIS as Future Vision for AI Assistants in AR

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Figure 1: An Iron Man inspired vision of a user interacting with a highly detailed holographic interface.

ABSTRACT

The fictional JARVIS assistant from the Iron Man movies depicts a compelling vision of computing: an intelligent agent seamlessly interwoven with immersive, spatial interfaces. In this position paper, we examine how current research trends in augmented reality and artificial intelligence are converging towards this vision, and identify five pillars that structure the path towards it: Systems Access, Agent Perception, Spatial Cognition, Agency, and Fluid Interaction. For each pillar, we survey the present state of research and outline current technological, ethical, and conceptual tensions as open challenges, providing a roadmap for turning this fiction into reality.

Index Terms: Augmented Reality, Agentic AI, Mixed-Initiative Interaction, Human-Agent Collaboration, Direct Manipulation, Spatial Computing, Situated Perception

1 INTRODUCTION

Interaction design is often driven by futuristic visions popularized in science fiction [8]. In this position paper, we explore how the design vision of JARVIS from “Iron Man” could become a reality: a holistic augmented reality (AR) environment, where artificial intelligence (AI) assistants and direct interaction are seamlessly intertwined (Fig. 1). In one scene, Tony Stark—a technological genius with a futuristic, holographic work environment—asks JARVIS—a close-to-omniscient AI personal assistant—to pull up an old blueprint. Stark then reaches into the blueprint himself to strip away the parts he does not need, isolates and rearranges what remains by hand, before handing the result back to JARVIS to work into something new. Delegation and direct manipulation happen on the same object, in the same breath, with no visible seam between

them. How can we move towards building such an AR system, embedding a powerful assistant, where delegation and direct control coexist without friction, and where do we stand?

Two dominant paradigms currently shape how such systems might actually be realized, and each captures only one half of that scene. Agentic AI systems handle complex, multi-step tasks well, drawing on broad context across a variety of objects and situations, but are comparatively clumsy at small, localized manipulation—rotating, nudging, or fine-positioning a single object in the dense, spatial context AR provides. Direct manipulation is the opposite: precise and immediate for exactly these local actions, but it does not scale to complex, multi-step tasks, which must still be carried out by hand, one action at a time.

Toward this vision, this position paper steps back from detailed research on individual components and instead focuses on the broader roadmap for realizing a system like JARVIS. From our research, we identify five pillars that, together, bridge agentic AI and direct manipulation rather than choosing between them: system and user act concurrently on the same environment, control shifting fluidly between the two rather than being granted or withheld. While a functional assistant might only need a subset of them, we believe the full vision rests on all five, summarized below and developed in full in the sections that follow.

- SA Systems Access** Embedded within the OS itself, running, reading, and manipulating any native application.
- AP Agent Perception** Continuous and total awareness of the scene, the system, and the user.
- SC Spatial Cognition** Precise spatial registration paired with human-legible intuition for where things belong.
- AG Agency** Control shifting fluidly between user and agent, without any handoff cost.
- FI Fluid Interaction** Multimodal, instinctive manipulation of objects.

2 VISION PILLARS

We describe the *what* and *why* of each pillar, provide a current *state of research*, and discuss potential *tensions* for each pillar. Rather than an exhaustive survey, we give a handful of examples to sketch the frontier where our pillars currently stand. An illustration of each pillar is provided in Fig. 2

2.1 SA Systems Access

What The assistant is embedded within the operating system itself, not deployed as another application; it can run, read, manipulate, display, and hide any application at will—regardless of how cooperative it is.

Why The content and capability a user actually relies on already lives inside mature, specialized applications—documents, maps, calendars, communication. An assistant that cannot reach into these applications is functionally blind to most of what the user does. Forcing users to abandon their existing tools in favor of new, assistant-native ones imposes a migration cost high enough to block adoption outright, and even where migration is free, a generalist reimplementation is rarely as good as any one task as the specialized tool it replaces. This is precisely the opposite of JARVIS, where every function is native to the assistant itself and no other software exists to interface with. Since an omnipotent AI remains out of reach, our vision keeps existing applications intact and asks the assistant to work with them, not to absorb them.

State of research Research into accessing existing applications is already underway. OpenClaw [1] connects agents directly to a machine’s filesystem, shell, browser, and credentials, and works such as OS-Symphony sustain long-horizon, multi-application workflows through a persistent memory of past steps [20]. Both work by scripting or controlling applications through interfaces that those applications already expose.

Tension This reliance on premeditated interfaces is also their limit, and it is one no single system can standardize away. Some applications expose structured interfaces through accessibility APIs or increasingly MCP-style¹ servers, while others expose nothing beyond a graphical user interface, leaving vision-based control as the only option. This is universal but brittle, where grounding failures on unfamiliar layouts can produce irreversible mistakes [15]. Standardization narrows this gap only for software that adopts it, leaving a long tail of unsupported applications that don’t. Whether a single agent can act reliably across this whole spectrum, not just the cooperative modern APIs, remains open.

2.2 AP Agent Perception

What Continuous, unified awareness of the scene, the system, and the user—expressed through voice, gesture, and gaze—used to resolve ambiguous, underspecified requests and anticipate upcoming needs.

Why Full context is often a hard requirement for completing a task, not merely a convenience: system understanding is needed to know what can be opened or acted on, scene understanding is needed to know what can be referenced or manipulated, and resolving what a gesture or glance refers to is only possible by fusing it with this same scene and system state. *Iron Man* seamlessly showcases all of that: sifting through old exposition models, Stark thinks aloud and gestures at half-finished structures rather than issuing clean commands, and JARVIS has to combine what is currently displayed, what Stark is pointing at among the scattered models, and the running thread of what he has been trying to reconstruct

¹The Model Context Protocol (MCP) is an open standard letting applications expose structured interfaces for agents.

across the session, to eventually surface the right piece of archive. Beyond enabling the agent to work at all, we believe perception is also necessary for low-friction interactions. Without it, every request must be explicitly and fully specified, and formulating such a request is itself a real cognitive cost, not just an inconvenience. Prior work extending Norman’s gulf of execution to LLM-based interaction shows that developing a fully specified prompt requires the user to mentally simulate a plan before ever executing it [16]—an effort we infer a strongly aware assistant could absorb on the user’s behalf.

State of research Fusing a small number of input channels to resolve ambiguity has already been demonstrated. MM-Conv [3] offers a large benchmark for resolving pronominal and partitive references from synchronized speech, gaze, and motion, while Bovo et al. pair an LLM with a segmented 3D scene and live application metadata to resolve deictic speech and pointing into concrete actions [2].

Tension The gap these leave is not feasibility but scope: no existing system unifies scene, system state, multimodal user input, and interaction history into a single interpretive layer. Even where several sources are fused, models tend to lean disproportionately on the easiest to process, typically text—thus disregarding the other modalities [19].

2.3 SC Spatial Cognition

What Precise geometric registration of virtual content in real-world coordinates, combined with human-like intuition for where things belong.

Why An assistant embedded in the OS and aware of context still cannot act in space without the ability to place, move, and register content precisely; without this, its promised reach never leaves the plane of language. *Iron Man* shows this precision at work: in a fight scene, JARVIS tracks an incoming missile and overlays a label directly on it, allowing both to find it at a glance, and to make available additional information. This is precisely the argument behind situated analytics: sensemaking is most effective when data and its physical referent stay closely connected in both space and time, rather than being pulled apart onto a separate screen [17]. And yet, placement without intuition undermines the value of delegating in the first place: occluding relevant real-world items, scattering related resources across disconnected areas of the workspace, or blocking the user’s line of sight forces the user to manually correct every result, silently handing the effort back to them.

State of research Recent work grounds virtual content directly in real, metric space rather than an abstract canvas. *Words into World* [5] lifts open-vocabulary detections into metric 3D anchors on commodity AR headsets, and Bovo et al. demonstrate the placement itself directly within a segmented scene [2]. On the presentation side, situated analytics has since been extended toward calmer, less obtrusive display, exploring how content tied to a physical referent can be surfaced without demanding constant attention [7].

Tension What existing systems achieve is spatial grounding: anchoring content to a detected surface or a specific referenced object. Thus far, this has only been demonstrated under simplified conditions, such as choosing among a small set of pre-identified candidate surfaces. Open-ended placement of new content anywhere in an arbitrary scene remains unsolved even at this easier, purely referential level. Beyond grounding and anchoring, judging where something should go, among several valid options—avoiding occlusion, grouping related items, respecting users’ habits—is an open problem for today’s models. Underlying both challenges—spatial grounding and judgment—is a mental model of space: a

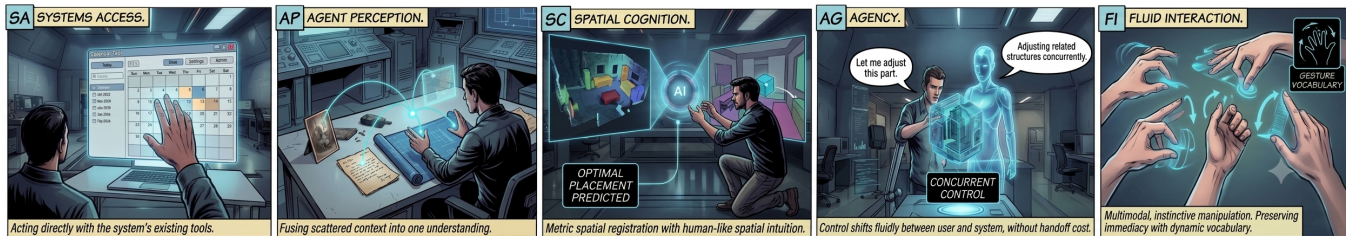


Figure 2: An illustration of our five pillars: Systems Access, Agent Perception, Spatial Cognition, Agency, and Fluid Interaction.

person can glance around a room and hold a coherent sense of its whole layout, including what lies just out of view, while current vision-language models, given those same partial glimpses, perform only marginally better than random guessing at recovering that layout [21]. Emerging world models, though still in their infancy, aim to internalize physical dynamics directly rather than infer them from 2D pixels [6], and could reshape this landscape within a few years.

2.4 AG Agency

What The user can seize or relinquish direct control of anything the system is acting on, at any moment: agency shifts fluidly between user and system, as in mixed-initiative interaction, never formally granted or returned.

Why Even with a powerful assistant, the user is still bound to an indirect paradigm of prompt-and-wait interaction. This is in sharp contrast to Iron Man’s system, allowing the user to interrupt JARVIS mid-edit and manipulate the blueprint by hand—all while talking to JARVIS for further refinement. To replicate this, the user must be able to seize what the system is doing as it unfolds, not only once it is done. This only works if the system enables the user to know what the agent is doing and to feel safe about interrupting. Indeed, if the agent is a black box, then no collaboration is possible—the user has no way of knowing how or when to interrupt. On the other hand, if the user feels like the interruption is likely to disrupt the process, they will be more hesitant to partake. Beyond seizing control, a strongly collaborative system also needs to balance how each agent—user and assistant—shares agency and contributes to the same goal.

State of research Non-turn-taking coordination between a human and an agent is only beginning to be studied directly. Collaborative Gym [14] moves in this direction with a shared workspace for tasks like web navigation and writing, letting a human intervene, redirect, or add requests without waiting on the agent to finish. Legibility for its own sake is also starting to draw attention: Kim et al. have recently made the case that LLM agents should proactively surface information about their own state, so users can build an accurate understanding of what the agent is doing [9].

Tension Collaborative Gym achieves take-over through explicit, discrete actions: a click, a typed message, a clear signal the system can key on. A physical takeover in AR offers no such event, only a continuous stream of motion, and detecting intent to take over from that alone remains unsolved. Even the non-AR case is not fully solved: Zou et al. find that interrupting agents to directly collaborate with them remains challenging even for the strongest current LLMs [22]. The reverse direction is just as open: for a system this broad, spanning apps, spatial placements, and background tasks at once, what minimal signal is actually enough to give a user genuine incentive to intervene remains unknown. Additionally, the problem may be structural rather than purely interactional: current long-horizon agent benchmarks explicitly treat human intervention

as a cost or bottleneck to minimize, rather than a resource to leverage [13]. This runs counter to what the kind of collaboration we advocate for actually requires, and might need an entirely different direction for how such models are trained or fine-tuned.

2.5 FI Fluid Interaction

What Direct manipulation with continuous, immediate visual feedback, smooth animated transitions, and no dead ends or breaks, so the user feels they’re directly touching and reshaping the interface rather than conversing with it [4].

Why Seizing control, as discussed in AG, only removes friction if what follows is itself fluid: control that lands on a clunky or unresponsive interface simply relocates the wait, from the assistant to the interaction itself. Any manipulation that first requires translating intent into a command for the system to interpret reintroduces exactly the mediation that seizing control was meant to escape; continuous representation of the object, physical action in place of commands, and immediate feedback are what remain once that mediation is stripped away [4]. This is precisely what makes the holographic interfaces of *Iron Man* feel as fluid as they do: gestures and pinches make information directly, physically manipulable, with nothing standing between the action and its effect.

State of research Individual techniques within this space continue to expand. Examples include PinchCatcher [10], which extends gaze-and-pinch interaction to multi-object selection, or Hand-Proxy [12], which lets a user control a virtual proxy hand through continuous natural speech, extending manipulation to situations where direct hand tracking is unavailable.

Tension Elmqvist et al.’s [4] own principles were drawn from 2D and tabletop exemplars, where the space of possible actions is comparatively small. The AR techniques above expand that space, but only one gesture or one input channel at a time; the space of possible embodied actions as a whole is far larger, and no closed set of “the” techniques covers it. When users are asked what gesture they would use for a given action, agreement holds for only a handful of canonical choices, with a long tail of idiosyncratic, person-specific ones beyond that [18]. Preserving this immediacy without being able to fully specify the vocabulary in advance remains open.

3 OPEN CHALLENGES

Across the examples discussed above, no single system spans more than a few pillars at once. Closing that gap is therefore not simply a matter of combining what already exists, but a research direction in itself. Beyond compatibility, realizing the vision as a whole raises new challenges that span across pillars, and that we believe also deserve attention from the community.

- **Privacy of total-scene access.** Eye tracking alone is already considered sensitive enough to uniquely identify a person from gaze patterns. A system built on SA and AP compounds this by design, continuously observing not just gaze but room geometry,

on-screen content across unrelated applications, hand movement, and habitual patterns, all at once. Securing, minimizing, or avoiding retention of information this comprehensive, without undermining the context the system depends on, is a direction worth pursuing in its own right.

This exposure is not limited to the user. Bystanders who happen to be in view have consented to none of it, and a system this continuous makes their presence part of a persistent record rather than a passing glance. Extending privacy protection to people who are not the user, but are nonetheless captured by it, is a related direction worth studying.

- **Centralization and data provenance.** A system this embedded does not just store data with a third party; it mediates nearly everything a person does, continuously. This is not simply a matter of infrastructure ownership: techniques like encryption or confidential computing can, in principle, let computation run on untrusted hardware without exposing the underlying data to whoever hosts it. The harder problem is provenance and explainability: whether a user can actually know, and verify, where their data goes, what is retained, and what is inferred from it, rather than merely being told. This opens two directions worth pursuing: keeping compute requirements low enough to run locally in the first place, or developing the trust infrastructure needed to run reliably on third-party hardware without exposure.
- **Dependence and loss of autonomy.** Studies already show that LLM assistance measurably lower neural, linguistic and behavioral engagement in writing [11]. A system as broad as the one we described stands to replicate, if not amplify those risks, to more topics. Studying this impact alongside the system's development, rather than after the fact, is a direction worth pursuing in its own right.

4 CONCLUSION

In this paper, we presented our vision for AR assistants that bridge agentic AI and direct manipulation. Inspired by Iron Man's JARVIS, we structured it around five pillars spanning systems access, agent perception, spatial cognition, agency, and fluid interaction. As the examples we discussed suggest, no existing system yet spans more than a fraction of these at once, and closing that gap raises open questions of privacy, deployment, and dependence we do not think can be addressed as an afterthought. We see this paper less as a finished proposal than as a starting point for the work, and the conversations, still ahead.

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Claude Sonnet 5 was used throughout the writing process to provide adversarial feedback and rephrasing. Figures were generated with Gemini.

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